LARAM School 2008: 1st week

	8.30	9.30	10.30	11.30	12.30	13.30	14.30	15.30	16.30	17.30
Mon 08/09	Introduction Bonnard, Cascini		break	I-1 Fell, Bonnard		Lunch	I-2 Cascini	break	I-3 Vardoulakis	
Tues 09/09	I-3 Vardoul.	I-4 (tutorial)	break	II-1 Fell		Lunch	II-1 Fell	break	II-2 Nadim	
Wed 10/09	II-3 Pastor		break	II-4 Bonnard		Lunch	II-5 Nadim		break	II-6 Fell
Thur 11/09	II-7 Bonnard		break	II-7 Bonnard		Lunch	II-8 Fell, Bonnard, Nadim		break	II-9 Fell
Fri 12/09	II-10 Pastor		break	II-10 Pastor		Lunch	III-1 student presentations		break	III-1 student pres.
Sat 13/09	Technical VISIT									

INTRODUCTION [1.5 hours]

(Bonnard, 0.5h): Introduction to LARAM 2008.

(Cascini, 1h): Introduction to landslides.

SESSION I [8 hours]

"Landslide analysis using approaches based on: Geology, Geotechnics and Geomechanics"

I-1 (Fell, Bonnard, 2h): Landslide identification. Key geological, geomorphological and hydrogeological features of: landslides in soils, large landslides and rock slides.

I-2 (Cascini, 1.5h): The geotechnical slope model.

I-3 (Vardoulakis, 3h): Basic geomechanics of landslides.

I-4 (Cascini, Fell, Vardoulakis, 1.5h): Tutorial.

SESSION II [22 hours]

"Risk Theory and Risk Analysis for Landslides"

II-1 (Fell, 2.5h): Landslide Risk Management concepts and framework and examples.

II-2 (Nadim, 2h): Deterministic and Probabilistic models for slope stability evaluation.

II-3 (Pastor, 2h): Introduction to modelling of catastrophic landslide events.

II-4 (Bonnard, 1.5h): Empirical models for travel distance.

II-5 (Nadim, 2h): Application examples of probabilistic methods and semi quantitative methods for landslide hazard zonation.

II-6 (Fell, 1.5h): Landslide Frequency Assessment.

II-7 (Bonnard, 3.5h): Different components of vulnerability to landslides. Prevention and long term management of landslides.

II-8 (Fell, Bonnard, Nadim, 2h): Case Studies: coal waste dump risk assessment (Fell), example from motorway in La Reunion Island (Bonnard), Aknes Rock slope in Norway (Nadim).

II-9 (Fell, 1.5h): Application of QRA to other geotechnical problems (Internal erosion of dams, crater lake hazard).

II-10 (Pastor, 3.5h): Advanced numerical models: initiation of landslides, propagation of sediments / climate change effects.

SESSION III [3.5 hours]

"Presentations by students"

III-1 (student presentations, 3.5h): Presentations by students about their research work + discussion

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LARAM School 2008: 2nd week

	8.30	9.30	10.30	11.30	12.30	13.30	14.30	15.30	16.30	17.30
Mon 15/09	IV-1 Cascini		break	IV-1 Cascini		Lunch	IV-2 Shiu		break	IV-2 Shiu
Tue 16/09	IV-2 Shiu		break	IV-3 van Westen		Lunch	IV-3 van Westen		break	IV-3 van Westen
Wed 17/09	V-1 Cascini	V-2 Nova	break	V-2 Nova		Lunch	V-3 Darve		break	V-3 Darve
Thur 18/09	V-4 Sorbino	V-5 Alonso	break	V-5 Alonso		Lunch	V-5 Alonso	break	VI-1 Leroi	
Fri 19/09	VI-1 Leroi		break	VI-2 Sorb	VI-3 Lacerda	Lunch	VI-3 Lacerda	break	VI-4 Sorbino	C. R.
Sat 20/09	Departures									

Session IV [14 hours]
"Landslide susceptibility, hazard and risk zoning at different scales"

IV-1 (Cascini, 4h): Input elements to zoning maps. Zoning scales, levels and methods. Basic methods and procedures for zoning at small and medium scales (<1:100,000 – 1:25,000). Tutorial on susceptibility and hazard zoning at 1:25,000 scale.

IV-2(Shiu, 4.5h): Natural terrain zoning and management criteria - Hong Kong practice and experience. Qualitative risk rating for individual slopes/hillsides and global quantitative risk assessment. Site-specific quantitative risk assessment and risk management. Tutorial on quantitative risk assessment.

IV-3(van Westen, 5.5h): GIS for landslide susceptibility and hazard assessment: spatial data and spatial analysis. Tutorial on landslide susceptibility assessment using statistical analysis.

Session V [12 hours]
"The role of sophisticated methods in landslide Risk analysis"

VI-1 (Cascini, 1h): Introduction to advanced slope stability characterization.

VI-2 (Nova, 3h): Analysis of the stability of soil slopes with low slope angles as a result of latent instability.

VI-2 (Darve, 3h): Finite element modelling of landslides by taking into account an hydromechanical coupling and an instability criterion.

VI-5 (Sorbino, 1hs): Flow-like mass movements in pyroclastic soils: triggering mechanisms and some remarks on propagation stage.

VI-2 (Alonso, 4h): Thermo-hydro-mechanical couplings in slope stability: the case of rapid drawdown, thermal effects in landslides. Tutorial on rapid slides.

SESSION VI [8 hours]
"Landslide risk management and mitigation"

VI-1 (Leroi, 4h): Risk management on la Désirade Island and Pointe-Noire in Guadeloupe.

VI-2 (Sorbino, 0.5h): Control works according to the risk mitigation framework.

VI-3 (Lacerda, 2.5h): Principles of prevention and long-term management of landslides, efficiency of drainage works. Case histories.

VI-4 (Sorbino, 1h): Case histories.

CONCLUDING REMARKS [1 hour]

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